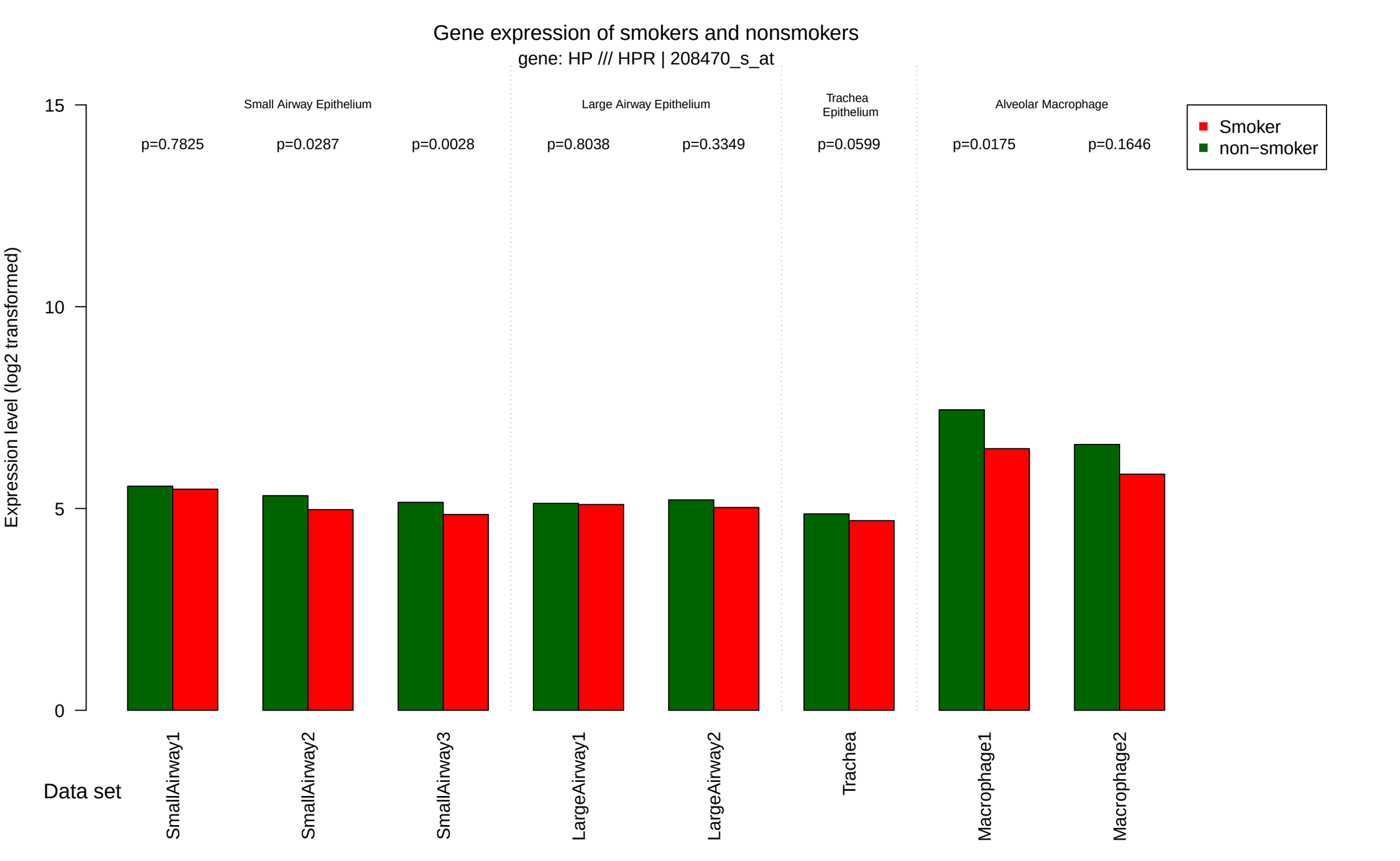
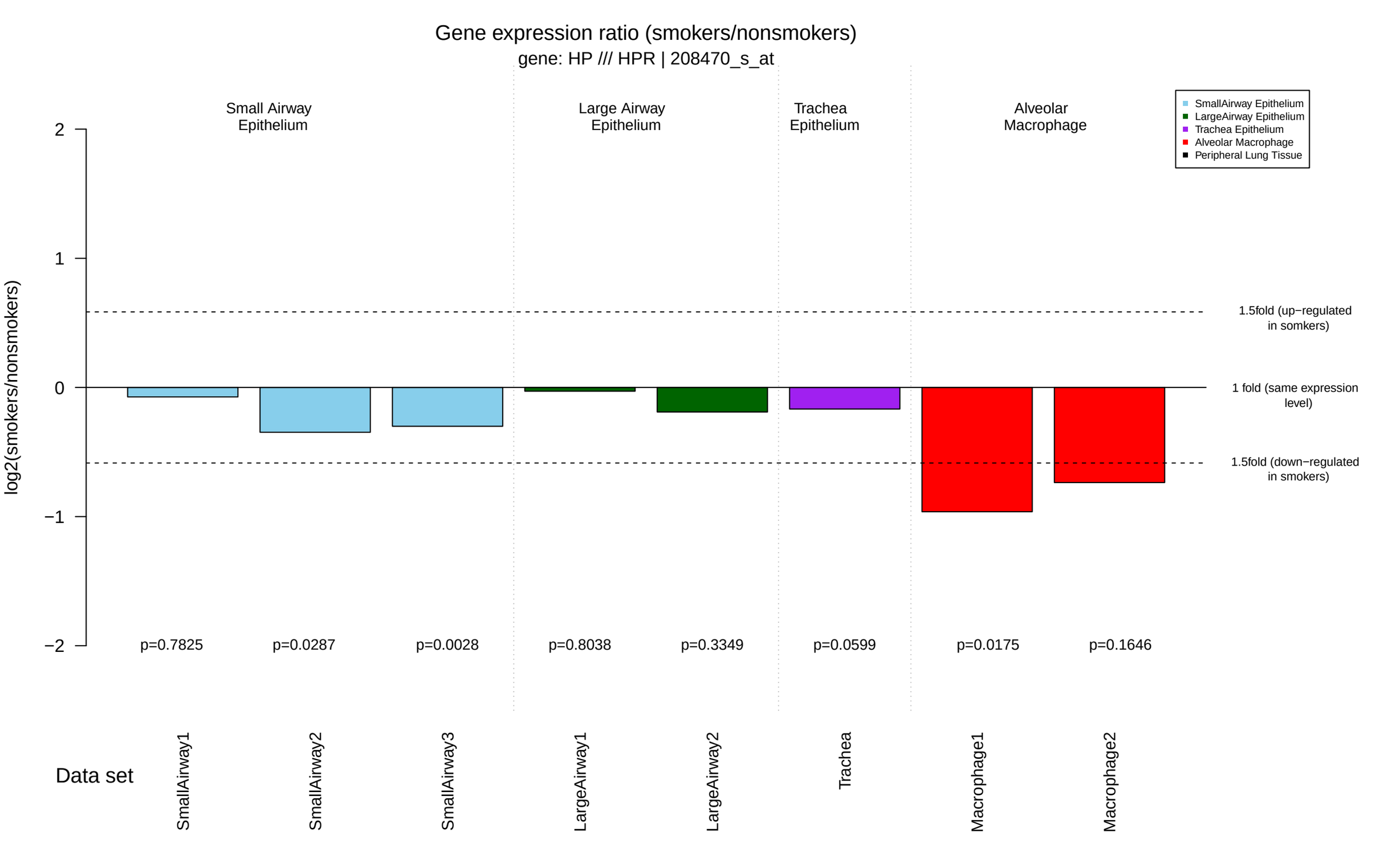
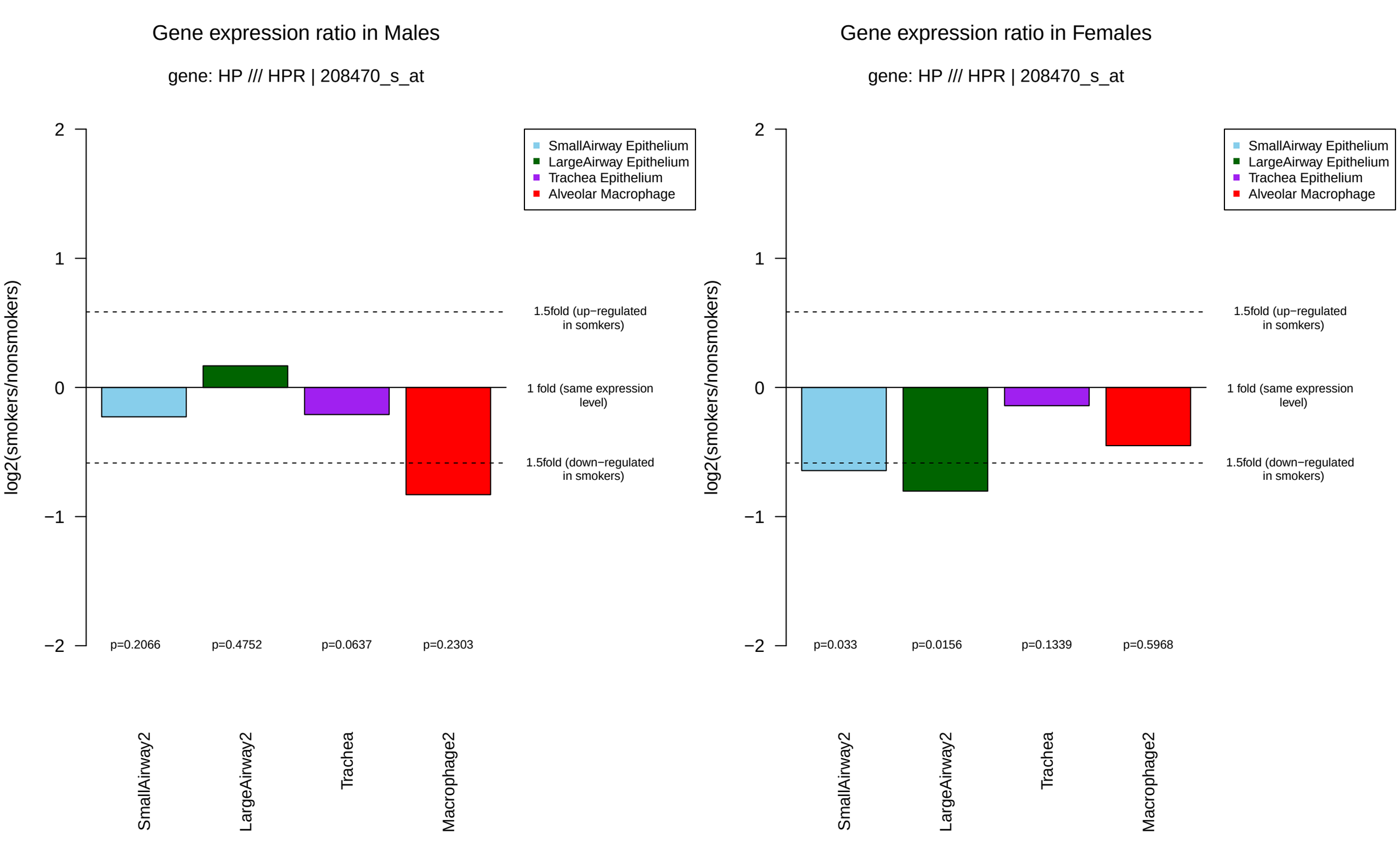
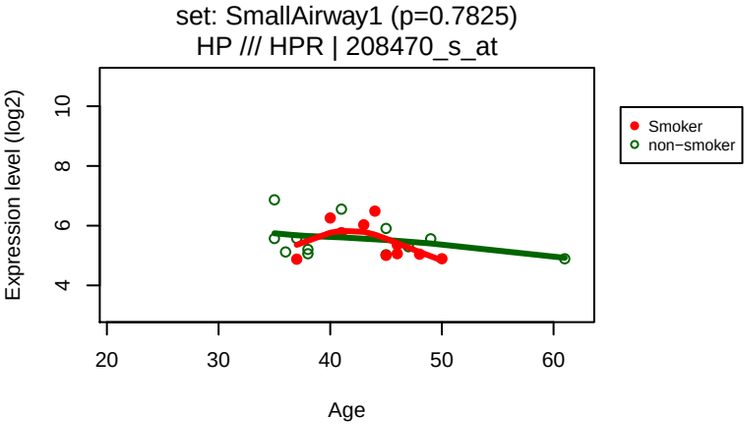
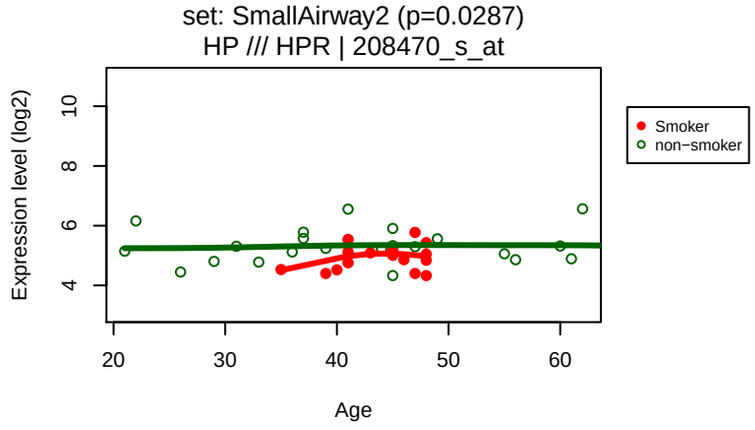
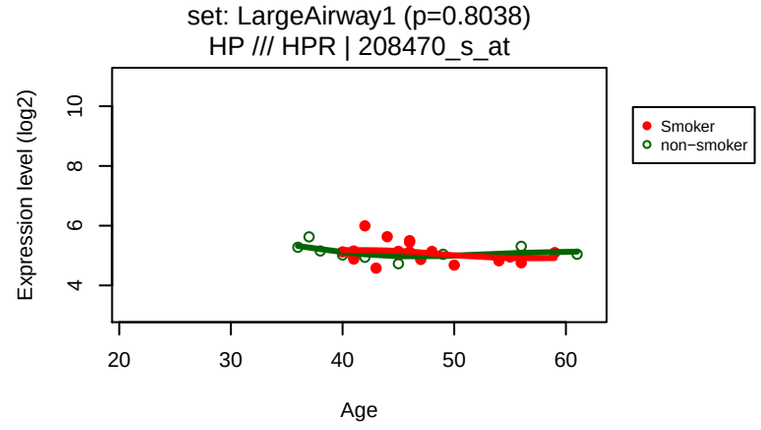

Gene expression of smokers and nonsmokers
gene: HP /// HPR | 208470_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7825
p=0.0287
p=0.0028
p=0.8038
p=0.3349
p=0.0599
p=0.0175
p=0.1646
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

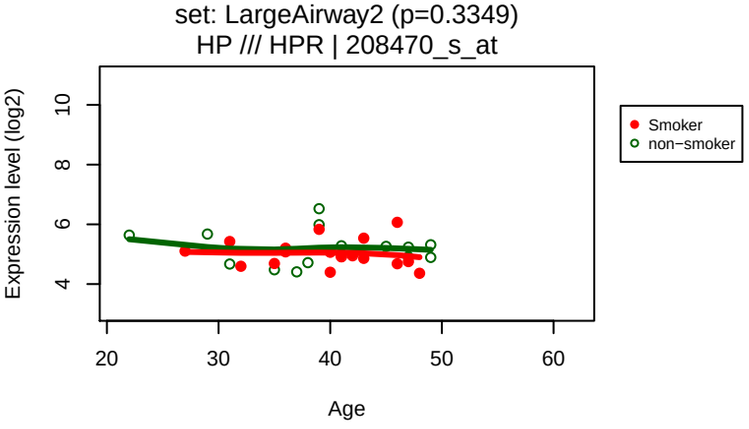

Gene expression ratio (smokers/nonsmokers)
gene: HP /// HPR | 208470_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7825
p=0.0287
p=0.0028
p=0.8038
p=0.3349
p=0.0599
p=0.0175
p=0.1646
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

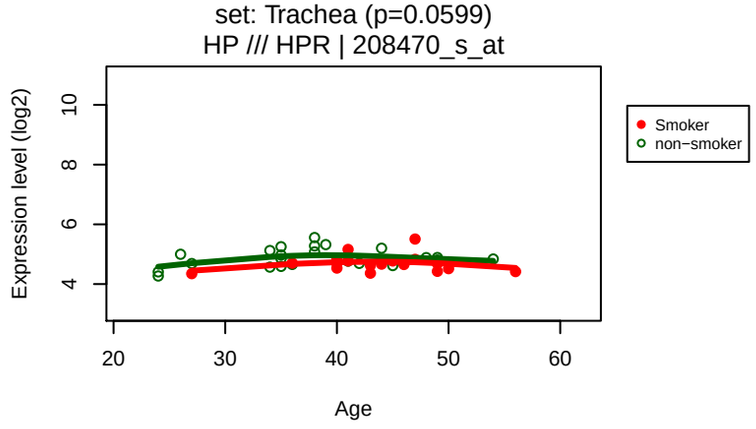

Gene expression ratio in Males
gene: HP /// HPR | 208470_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2066
p=0.4752
p=0.0637
p=0.2303
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HP /// HPR | 208470_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.033
p=0.0156
p=0.1339
p=0.5968
SmallAirway2
LargeAirway2
Trachea
Macrophage2

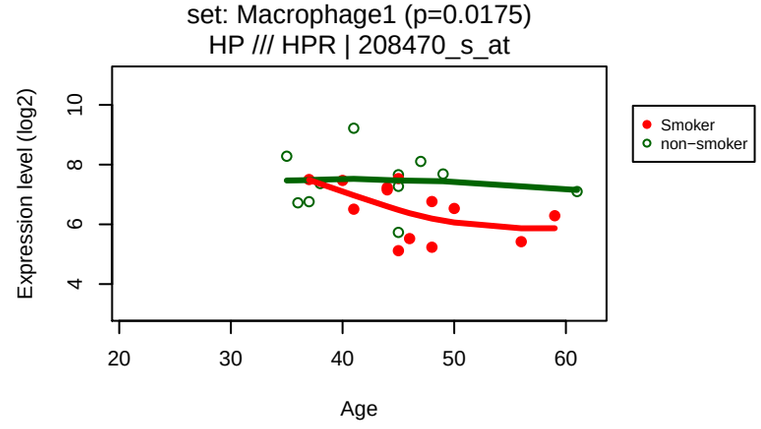

set: SmallAirway1 (p=0.7825)
HP /// HPR | 208470_s_at
Expression level (log2)
Age
Smoker
non−smoker

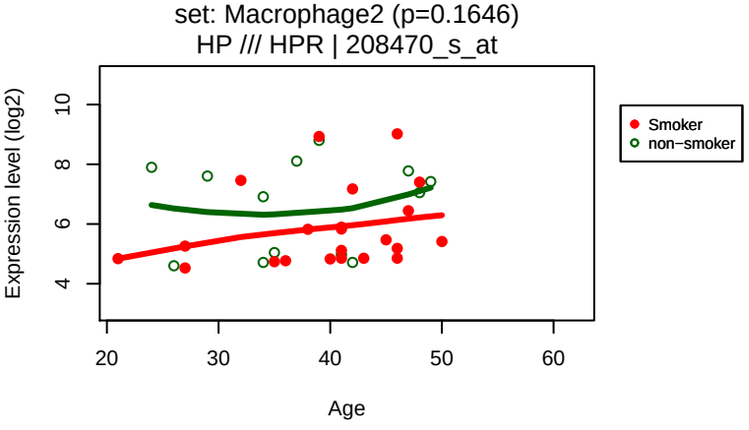

set: SmallAirway2 (p=0.0287)
HP /// HPR | 208470_s_at
Expression level (log2)
Age
Smoker
non−smoker

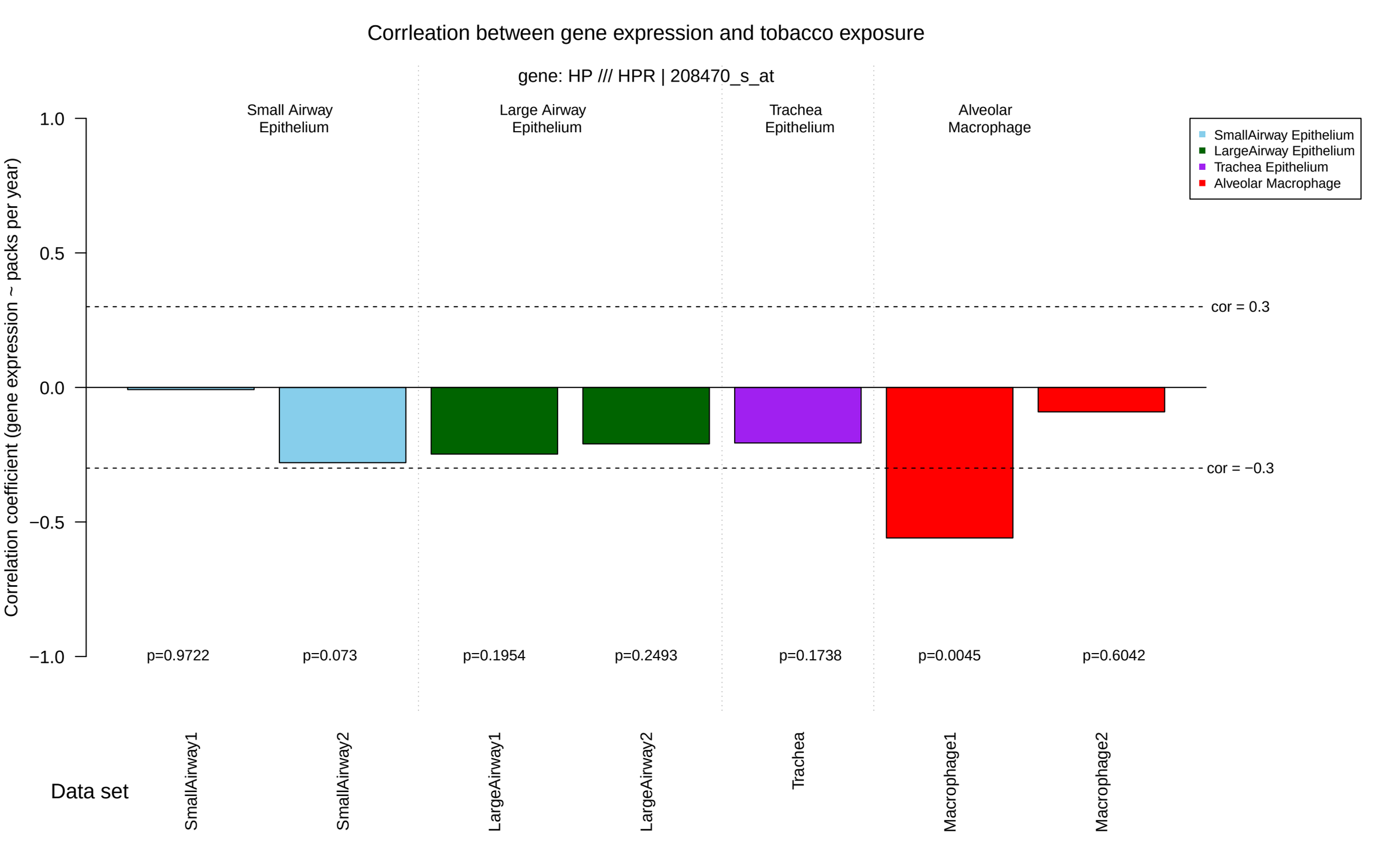

set: LargeAirway1 (p=0.8038)
HP /// HPR | 208470_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3349)
HP /// HPR | 208470_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0599)
HP /// HPR | 208470_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0175)
HP /// HPR | 208470_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1646)
HP /// HPR | 208470_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HP /// HPR | 208470_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9722
p=0.073
p=0.1954
p=0.2493
p=0.1738
p=0.0045
p=0.6042
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2